

CS & IT ENGINEERING



Computer Network

Error Control

Lecture No. - 04



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Recap of Previous Lecture



Topic

CRC



Topics to be Covered



Topic

CRC

ABOUT ME



Hello, I'm **Abhishek**

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#Q. Consider the message $M = 1010001101$. The cyclic redundancy check (CRC) for this message using the divisor polynomial $x^5 + x^4 + x^2 + 1$ is

[GATE 2005]

✓ (A) 01110

(B) 01011

(C) 10101

(D) 10110

Ans: A

$$\begin{array}{r}
 110101 \overline{) 1010001101000000} \\
 \underline{110101} \\
 111011 \\
 \underline{110101} \\
 111010 \\
 \underline{110101} \\
 111110 \\
 \underline{110101} \\
 101100 \\
 \underline{110101} \\
 110010 \\
 \underline{110101} \\
 01110
 \end{array}$$

CRC

$$G(x) = x^5 + x^4 + x^2 + 1$$

Divisor = 110101

#Q. The message 11001001 is to be transmitted using the CRC polynomial $x^3 + 1$ to protect it from errors. The message that should be transmitted is :

[GATE 2007]

(A) 11001001000

✓ (B) 11001001011

~~(C) 11001010~~

~~(D) 110010010011~~

Ans: B

$$\begin{array}{r}
 1001 \overline{) 11001001000} \\
 \underline{1001} \\
 1011 \\
 \underline{1001} \\
 1000 \\
 \underline{1001} \\
 1100 \\
 \underline{1001} \\
 1010 \\
 \underline{1001} \\
 011 \\
 \underbrace{}_{CRC}
 \end{array}$$

$$G(x) = x^3 + 1$$

$$\text{Divisor} = 1001$$

#Q. A computer network uses polynomials over $GF(2)$ for error checking with 8 bits as information bits and uses $x^3 + x + 1$ as the generator polynomial to generate the check bits. In this network, the message 01011011 is transmitted as :

[GATE 2017]

(A) 01011011010

(B) 01011011011

✓ (C) 01011011101

(D) 01011011100

Ans: C

$$\begin{array}{r}
 1011 \overline{) 01011011000} \\
 \underline{1011} \\
 1100 \\
 \underline{1011} \\
 1110 \\
 \underline{1011} \\
 101 \\
 \text{CRC}
 \end{array}$$

$$G(x) = x^3 + x + 1$$

$$\text{Divisor} = 1011$$

#Q. Consider the cyclic redundancy check (CRC) based error detecting scheme having the generator polynomial X^3+X+1 . Suppose the message $m_4m_3m_2m_1m_0 = 11000$ is to be transmitted. Check bits $c_2c_1c_0$ are appended at the end of the message by the transmitter using the above CRC scheme. The transmitted bit string is denoted by $m_4m_3m_2m_1m_0c_2c_1c_0$. The value of the checkbit sequence $c_2c_1c_0$ is :

[GATE 2021, Set-2, 2-Mark]

(A) 101

(B) 110

✓ (C) 100

(D) 111

Ans: C

$$\begin{array}{r}
 1011 \overline{) 110000000} \\
 \underline{1011} \\
 1110 \\
 \underline{1011} \\
 1010 \\
 \underline{1011} \\
 100
 \end{array}$$

CRC

$$G(x) = x^3 + x + 1$$

$$\text{Divisor} = 1011$$



Topic : CRC

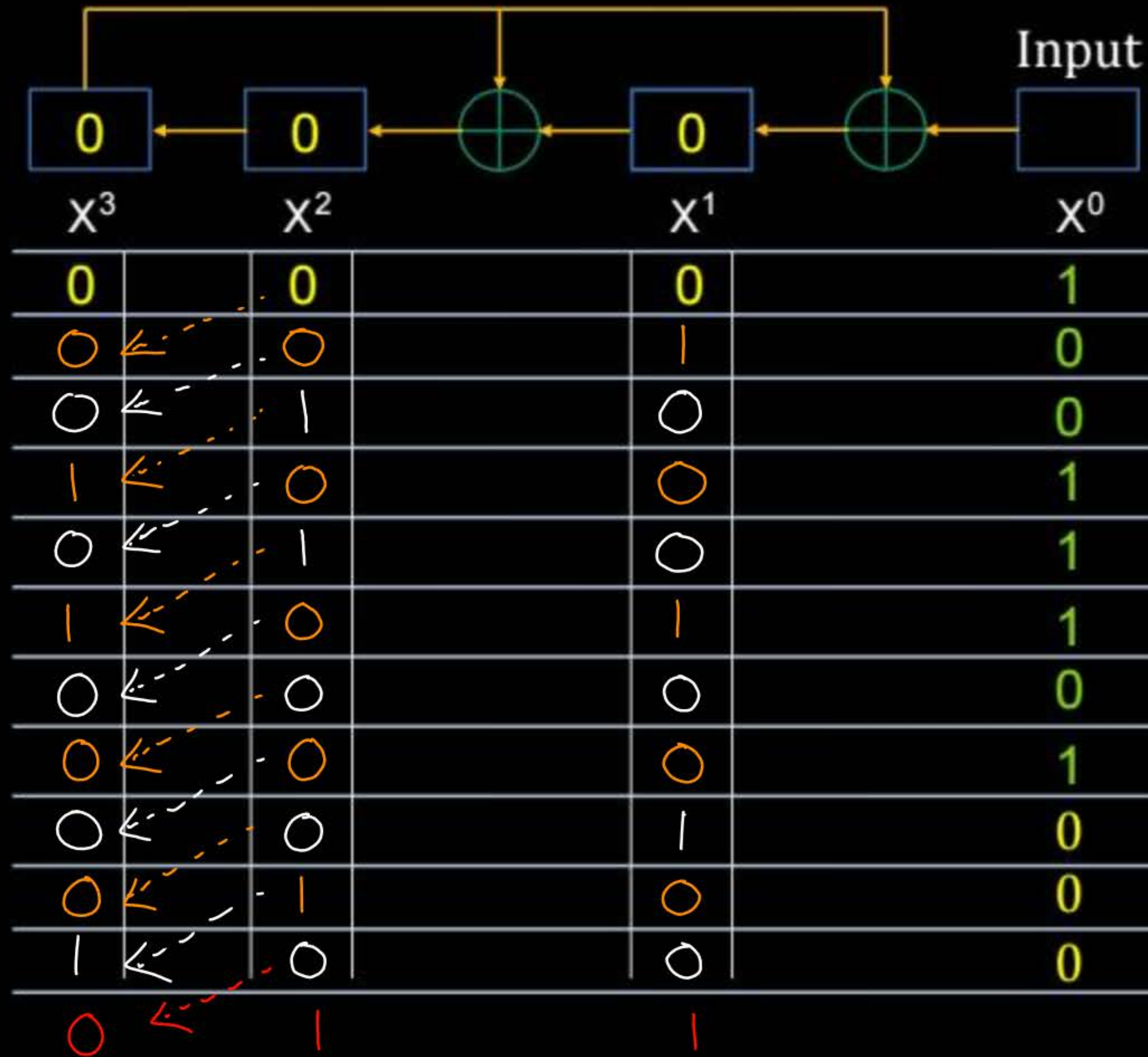


Example 2:

$$\underline{G(X)} = X^3 + X + 1$$

$$\text{Message (DATA)} = 10011101$$

$$\underline{\text{CRC}} = 011$$

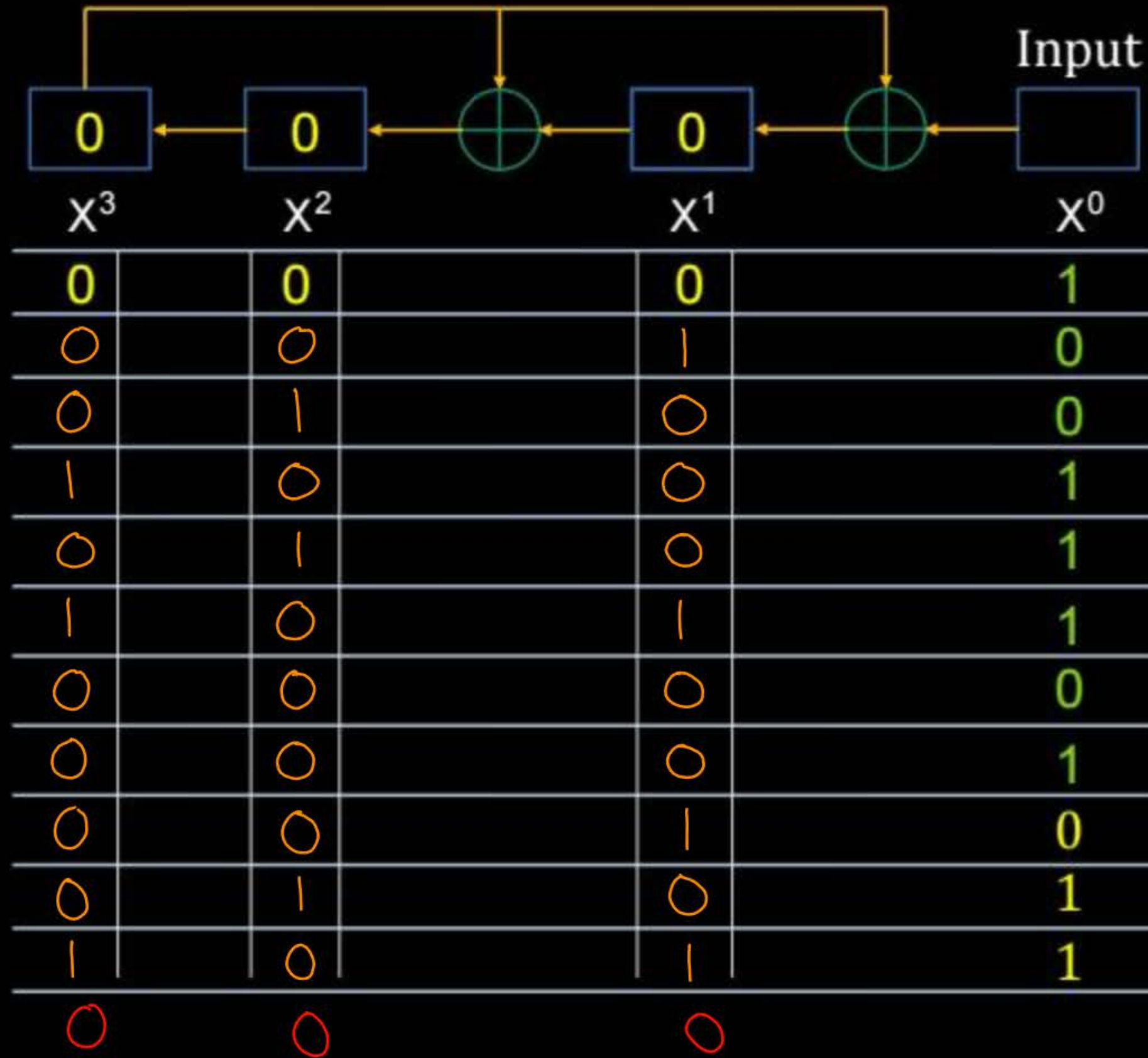


AT Sender (Transmitter)

Input = 1 0 0 1 1 1 0 1 0 0 0

DATA 3 zeros

CRC = 011



AT Receiver

Input = 1 0 0 1 1 1 0 1 0 1 1

DATA CRC

if $R'(X) == \text{ZERO}$:
 then Receiver concluded
 “No any error detected”
 else
 Receiver concluded
 “Error detected”



Topic : CRC



CASE I : No any error

Transmitter transmit : $[M(X) * X^n] + [R(X)]$

Receiver received : $[M(X) * X^n] + [R(X)]$

Receiver protocol :

$[M(X) * X^n + R(X)]$ [Modulo-2 Division] $[G(X)]$

Above equation will definitely lead “zero remainder” ✓

Receiver conclude : “No any error detected”

$$E(X) = 0$$



Topic : CRC



CASE II : Error Included

Transmitter transmit : $[M(X) * X^n] + [R(X)]$

Receiver received : $[M(X) * X^n] + [R(X)] + [E(X)]$

$$E(X) \neq 0$$



Topic : Error Polynomial



E(X) : Error Polynomial Function

→ Coefficient are either Zero or One

Data : m bits CRC : n bits

Codeword : (m + n) bits

Degree(E(X)) < (m + n) ✓



Topic : Error Polynomial



For single bit error : $\rightarrow E(x)$ contains only one term

$$E(X) = X^i$$

Where $[0 \leq i < (m + n)]$



Topic : Error Polynomial



Transmitter transmitted :

$$X^{10} + X^7 + X^6 + X^5 + \underline{X^3} + X + 1$$

1 0 0 1 1 1 0 1 0 1 1



Receiver received :

$$X^{10} + X^7 + X^6 + X^5 + X + 1$$

1 0 0 1 1 1 0 0 0 1 1

③



$$E(X) = X^3$$



Topic : Error Polynomial



Transmitter transmitted :

$$X^{10} + X^7 + X^6 + X^5 + X^3 + X + 1$$

1 0 0 1 1 1 0 1 0 1 1

Receiver received :

$$X^{10} + X^7 + X^6 + X^5 + \underline{X^4} + X^3 + X + 1$$

1 0 0 1 1 1 1 1 0 1 1

$$E(X) = \underline{X^4}$$



Topic : Error Polynomial



For two bit error :

$E(x)$ contains two terms

$$E(X) = (X^i + X^j)$$

Where i & j are 0 to $(m + n - 1)$ and $i > j$

$$\text{Burst length} = (i - j + 1)$$



Topic : Error Polynomial



Transmitter transmitted :

$$X^{10} + X^7 + X^6 + X^5 + X^3 + \underline{X} + 1$$

1 0 0 1 1 1 0 1 0 1 1



Receiver received :

$$X^{10} + \underline{X^9} + X^7 + X^6 + X^5 + X^3 + 1$$

1 1 0 1 1 1 0 1 0 0 1



$$E(X) = \underline{X^9} + \underline{X}$$



Topic : CRC



CASE II : Error Included

Receiver protocol :

$$[M(X) * X^n + R(X) + E(X)] \text{ [Modulo-2 Division] } [G(X)]$$

$$\frac{(A+B+C)}{G} = \frac{(A+B)}{G} + \frac{C}{G}$$



Topic : CRC



CASE II : Error Included

Receiver protocol :

$$\begin{aligned} & \boxed{[M(X) * X^n + R(X)] \text{ [Modulo-2 Division] } [G(X)]} \Rightarrow \text{This division always} \\ & \text{lead zero remainder} \\ & + \boxed{[E(X)] \text{ [Modulo-2 Division] } [G(X)]} \Rightarrow \text{Remainder of this division} \\ & \text{is } R'(x) \end{aligned}$$

if $E(x)$ is zero
then $R'(x)$ is definitely zero.

if $G(x)$ completely divide $E(x)$
then $R'(x)$ is zero.

else

$R'(x)$ is non-zero

int A, B

if $(A < B)$

then $(\frac{A}{B})$ always lead non-zero remainder

if $(A \geq B)$

then $(\frac{A}{B})$ may lead non-zero remainder.



Topic : CRC



CASE II : Error Included

Receiver protocol :

$$[E(X)] \text{ [Modulo-2 Division]} [G(X)]$$

→ $G(X)$ have $(n+1)$ terms (length)

$$G(X) = x^n + \dots + 1$$

→ If $E(X)$ causes at-most n length burst error
then above equation always lead non-zero remainder



Topic : CRC



Example 3 :

$$G(X) = X^3 + 1, \text{ Divisor} = 1001$$

$$\text{Data} = 10011101 \quad \text{CRC} = ? = 100$$

$$\text{Transmitted Data} = \underbrace{10011101}_{\text{DATA}} \underbrace{100}_{\text{CRC}}$$



Topic : CRC



Example 3 :

$$\underline{G(X)} = X^3 + 1$$

$$\text{Transmitted Data} = 10011101100$$

$$\underline{E(X)} = X^6 + X^3$$

$$\text{Received Data} = 10010100100$$

$$\frac{E(X)}{G(X)} = \frac{(X^6 + X^3)}{(X^3 + 1)}$$

$$= \frac{X^3 \cdot (X^3 + 1)}{(X^3 + 1)}$$

$$= X^3$$

(Complete division)

$$\begin{aligned} \text{Burst length} &= (6 - 3 + 1) \\ &= 4 \end{aligned}$$



Example 3 :

$$G(X) = X^3 + 1$$

Received Data = 1 0 0 1 0 1 0 0 1 0 0

Receiver Conclusion : No error detected
[$R'(x)$ is zero]

$$\begin{array}{r} 1001 \overline{) 100101001000} \\ \underline{1001} \\ 1001 \\ \underline{1001} \\ 0 \end{array}$$



Topic : CRC



Example 4 :

$$G(X) = X^3 + 1$$

$$\text{Transmitted Data} = 10011101100$$

$$E(X) = X^5 + X^3$$

$$\text{Received Data} = 10011000100$$

$$\frac{E(X)}{G(X)} = \frac{(X^5 + X^3)}{(X^3 + 1)}$$

$$= \frac{X^3 + (X^2 + 1)}{X^3 + 1}$$

(Not complete division)

$$\begin{aligned} \text{Burst Length} &= (5 - 3 + 1) \\ &= 3 \end{aligned}$$



Topic : CRC Property ✓



→ CRC can detect any length burst error,
up-to the degree of generator polynomial function ✓



2 mins Summary



Topic

CRC



THANK - YOU

